Tianrui Qiu

Mechanical & Electrical Engineering Student | CAD | Robotics | MATLAB

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PROFILE

I am a first-year Mechanical and Electrical Engineering student at the University of Bristol, based in Bristol and Shanghai Zhangjiang during breaks. I learn fast, enjoy turning ideas into physical prototypes, and focus on CAD modelling, robotics, MATLAB analysis, and practical AI tools. Full project evidence is available at tianruiqiu.com.

EDUCATION

University	University of Bristol
Degree	BEng Mechanical and Electrical Engineering, currently studying
Period	Sep 2025 - Present

INTERNSHIP TARGET

Open to 2026 summer internships in mechatronics, medical robotics, vehicle electronic chassis, product engineering, or AI Agent workflows.

SELF-DRIVEN LEARNING

- **Project-backed self drive:** I do not want self motivation to stay as a claim. My two open-source AI Agent projects are evidence that I turn repeated learning problems into usable tools.
- **AI plus engineering workflow:** *video-transcribe-skill* uses Python, FFmpeg, and local Whisper to turn unsubtitled tutorials or demos into transcripts that Codex can summarize, extract from, and answer questions about.
- **Method discipline:** *codex-agent-custom-instructions* documents how I use coding agents: clarify first, make small edits, keep deterministic steps in scripts, and verify the result.

TECHNICAL SKILLS

CAD & Design	Programming & Analysis	Prototyping
Fusion 360, assembly modelling, technical drawings, mechanical drawing	Python, MATLAB, parametric modelling, heat-sink calculation	3D printing, mechanical assembly, packaging checks, arena/lab testing

PROJECTS

AI Agent / Codex Skills Open-Source Projects
2026 | AI Agent / Python

- Built *video-transcribe-skill* for a real engineering learning problem: many useful mechanical, robotics, and software tutorials have no subtitles, so an AI agent needs transcript evidence before it can summarize or answer questions reliably.
- Used Python to connect video source resolution, FFmpeg audio processing, local faster-whisper transcription, and transcript/metadata output, turning video-based learning into a repeatable workflow.
- Separated deterministic tooling from model judgment: scripts handle download, routing, conversion, and file checks; Codex reads the transcript for summaries, extraction, and Q&A.
- Open-sourced *codex-agent-custom-instructions* as a bilingual checklist for using coding agents with fewer guesses, smaller edits, clearer failures, and more verifiable outputs.

First-Year Design Portfolio
2025-2026 | Design / CAD

- Built a first-year design portfolio from hand construction to Fusion 360 modelling, software technical drawings, a 14-item parts list, and an interactive GLB model.
- Practised engineering drawing discipline through views, dimensions, tolerances, section/detail drawings, callouts, and reviewable documentation.
- Converted the exported assembly into a browser-ready GLB model and kept the Fusion 360 source package as supporting portfolio evidence.

Robot War Society Project 2025-2026 | Robotics / Manufacturing

- Worked on a University of Bristol BEEES Robot Wars compact flipper robot, covering model development, 3D-printed structure, physical assembly, and competition preparation.
- Handled packaging constraints around drive, battery, ESC, receiver, wiring, shell geometry, fastener access, and serviceability before arena testing.
- Used physical assembly and arena feedback to identify CAD issues around clearance, wall thickness, screw access, reliability, repairability, and next iteration priorities.

High-Power LED Cooling Analysis 2025-2026 | Thermal / MATLAB

- Built a fan-coupled MATLAB heat-sink model from report equations, including fan curve interpolation, operating-point calculation, and thermal resistance output.
- Compared predictions against 25/35/45 mm lab data and linked cooling behaviour to LED system efficiency.
- Reviewed error sources including airflow assumptions, contact resistance, material simplifications, and experimental measurement uncertainty.